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Image encryption using DNA coding and hyperchaotic system

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ABSTRACT

The paper proposes a grey scale image encryption scheme using six-dimensional Lorenz chaotic system. The key generating system uses a 32 character (256 bits) key as input. The system uses simple modular arithmetic and bitwise XOR operations followed by coupling a series of Lorenz six-dimensional systems to generate the key-stream. This generated key stream is used in the encryption and decryption algorithm of the plain image. The basic permutation and diffusion architecture, using the fundamental properties of modular arithmetic, bitwise XOR operations and DNA encoding/decoding technique, are executed to confuse the relationship between plain image and the ciphered image. The experimental results and security tests confirm that the proposed image encryption algorithm possess high security and can be a good candidate for practical image encryption.

KEYWORDS

Lorenz system; chaos; encryption; decryption; DNA coding; information entropy; NPCR; UACI

1. INTRODUCTION

Now-a-days data transmission is prominent in any sort of communication and the importance of a particular type of data can vary from lower to higher level. In the present age of technology, cybercrime is quite common and the security of data may easily be compromised. In order to counter the unauthorised access and data theft, a suitable data protection technique should be developed. Data encryption is one of the important data protection techniques which widely used in information security. Image data encryption has gained popularity since any type of data can be represented in the form of an image. Conventional cryptographic algorithms like AES, DES, may be quite successful in encrypting raw data of small size but cannot provide good results in encryption of the digital images. These algorithms are unable to exploit the correlation among the adjacent pixels, which is an inherent property of an image. Rigorous research is going on the techniques of encrypting an image. Any standard encryption algorithm performs basic three operations: pixel permutation, diffusion and confusion. However, as the time passed by, these encryption algorithms started to fall prey to certain stringent security test. Desirable randomness in the generated sequence could not be obtained from these algorithms. Hence non-linear science started to gain popularity. Chaos theory introduced by Edward Lorenz became the building block of chaos based image encryption techniques. A certain set of deterministic equations composed of a number of state

variables and fixed parameters may result in a sequence which is deterministic but unpredictable when iterated for a number of times. The generated sequence exhibits an inherent behavior of randomness and this property is widely exploited in designing the chaos based cryptosystems. In the literature of chaos-based cryptography, generally, four types of chaotic systems (discrete time, continuous time, time delay and hyperchaotic) used as entropy sources in an encryption algorithm. The role of the chaotic system in encryption algorithm is to generate the chaotic key sequence for pixel permutation and confusion. A few examples of chaos based cryptosystems along with their merits and demerits are discussed in the next section.

2. RELATED LITERATURE REVIEW

Hong *et al.* have proposed a suitable chaos based using two round encryption with two different chaotic sequences generated from the logistic map and hyperchaotic system, respectively, in ref. [1]. Min *et al.* combined a piecewise linear chaotic map with a logistic map in reference [2] to produce a chaotic pseudo random number sequence and applied encrypting image. Lin and Wang have proposed an chaotic image encryption method by using the basic process of image scrambling, pixel replacement and piecewise linear memristor in Chua's circuit in reference [3]. An image encryption algorithm using chaotic Baker map and Lorenz system is

introduced in reference [4]. It proposes an image encryption algorithm in which the key stream is dependent on the input plain image. Further, the state variables of the Lorenz system are selected according to the pixel values of the plain image. Reference [5] proposes an image encryption technique with orthogonal polynomial transformation in the frequency domain and bit shuffling using tent map in the spatial domain. In reference [6] an image encryption algorithm employing Latin rectangle scrambling using 2D chaotic Henon map and bi-directional pixel diffusion using piecewise linear chaotic map is proposed. The algorithm claims to reach high security in only two rounds. A novel double image encryption technique using Blowfish algorithm and cross chaos map is presented in reference [7]. A dynamical image encryption algorithm based on parameter varying logistic chaotic map is presented in reference [8]. Reference [9] has presented a new block based image encryption. It proposes to use cat map as well as logistic map in encryption process. The input image dependent key stream generated from these maps are used in the diffusion process. The algorithm claims secured enough in practical transmission and reception. A colour image encryption algorithm using 4D Lorenz system is reported from our group in an international conference [10]. In this paper, the colour image is split into three R G B channel and they are encrypted by the mechanism of pixel diffusion and confusion with the help of chaotic key generated from 4D Lorenz system. Next pixel permutation is done among three colour components with the help of chaotic quadratic and cubic maps. In the above mentioned references, the encryption algorithms have involved two round operations and have used double or multiple chaotic systems for chaotic key generation. Since a two round encryption takes longer execution time in comparison to a single round encryption and the inclusion of double or multiple chaotic systems increases the system complexity, we need to develop a fast and highly secured chaotic encryption system with relatively less system complexity.

Recently, chaos based DNA coding and DNA addition-subtraction arithmetic are being used in several image encryption algorithms [11–15] to resist the known-plaintext, brute force, differential and chosen-plaintext attacks. In reference [11] an image encryption algorithm utilizing chaotic systems, DNA encoding is presented. In this algorithm, the logistic system is iterated to produce the DNA matrix and set of encoding/decoding rules is framed. DNA addition operation is performed using the generated DNA matrix and the DNA encoded image. The DNA encoded image diffusion is done through the iterations of the logistic system to increase the sensitivity and

information entropy of the cryptosystem. A spatiotemporal chaos based on coupled map lattice is used in ref. [12] to generate chaotic sequence and pixel confusion by XOR operation. Then DNA encoding followed by position permutation is done. Then a specific rule based DNA decoding is performed to generate the ciphered image. Other noteworthy algorithms are discussed in references [13–17]. Some of the algorithms stated above do not give good results in various security analysis or not technically sound in the view point of computational complexity and execution time.

Therefore, the aim of this paper is to propose a new encryption system that can reduce the complexity of the encryption algorithm by considering only one hyperchaotic system for pixel confusion, permutation, DNA encoding, complementing and decoding to achieve the ciphered image with higher security. In this algorithm, a six-dimensional chaotic Lorenz system is used to design a cryptosystem having reduced complexity and execution time compared to the reported work in the literature recently. A key matrix is generated by taking a 32 character secret key as input. The generated key matrix is reshaped into a matrix with the same dimension of the input image. The pixel positions of the plain image are randomized using chaotic key matrix. After XOR operation between plain image and chaotic key, DNA encoding, complement shifting and DNA decoding are applied sequentially to execute the encryption process. The rest of the paper is organized as follows: Section 3 outlines the 6D Lorenz chaotic system and its relevant characteristics. The basic rules for DNA encoding and decoding techniques are illustrated in Section 4 along with suitable examples. Section 5 presents the grey scale image encryption and decryption algorithms. Section 6 presents the experimental results and the security analysis of the proposed cryptosystem and the article is concluded in section 7.

3. LORENZ CHAOTIC SYSTEM

The proposed image cryptosystem uses a hyperchaotic six-dimensional Lorenz system [18] as stated below:

$$\left. \begin{aligned} \dot{x} &= a(y - x) + u \\ \dot{y} &= cx - xz - y + v \\ \dot{z} &= -bz + xy \\ \dot{u} &= du - xz \\ \dot{v} &= -ky \\ \dot{w} &= hw + ly \end{aligned} \right\} \quad (1)$$

where x, y, z, u, v, w are the state variables and a, b, c, d, k, h and l are the parameters. In our algorithm we have taken $a = 10, b = 8/3, c = 28, d = 2, k = 8.4, h = 1$ and $l = 1$

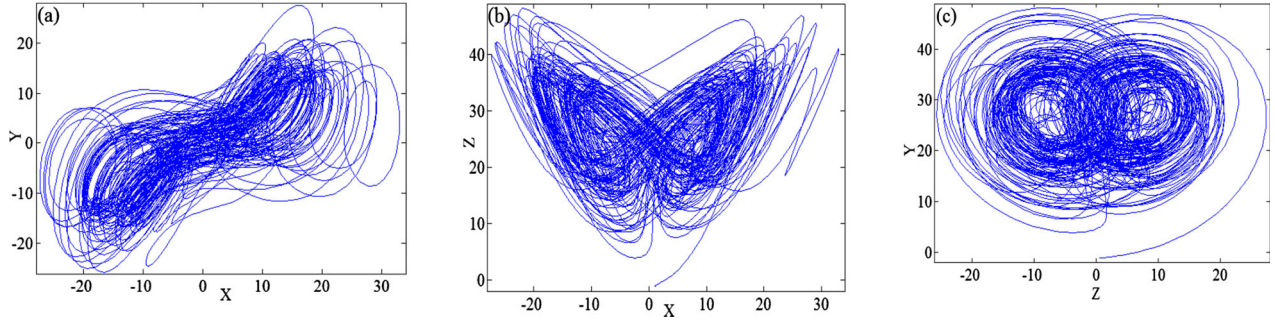


Figure 1: The projections of attractors of 6D Lorentz system. (a) In the x–y space, (b) In the x–z space, (c) In the z–y space.

as the parameter values. The projection of attractors is shown in Figure 1. The 6D chaotic Lorentz system has been chosen to design the cryptosystem since it exhibits the property of randomness with higher degree of unpredictability. Its higher chaotic nature has made it better suitable candidate for secure communication applications. The characteristic properties of this hyperchaotic system are: (a) it is a six-dimensional autonomous system, (b) It has a simple algebraic structure with complex dynamical behaviour, (c) The six Lyapunov exponents of the system are, $\lambda_1 = -2.667$, $\lambda_2 = -22.6858$, $\lambda_3 = 0.3260$, $\lambda_4 = 1$, $\lambda_5 = 2$ and $\lambda_6 = 11.3599$. Since it has four positive Lyapunov exponents and the sum of all the Lyapunov exponents are less than zero it can produce stable but unpredictable orbits which implies the generation of chaos.

4. THE DNA ENCODING AND DECODING PROCESS

The Deoxyribonucleic acid (DNA) is a molecule that contains the genetic information of all the living organisms. The development and function of the organisms depend on the structure of its DNA molecule. A DNA structure consists of four nucleic acid bases adenine (A), cytosine (C), guanine (G) and thymine (T). According to Watson-Crick base pairing rules [11,19], A and T are complementary and G and C are complementary. In the binary representation 0 and 1 are complementary, similarly, we can say 00 and 11 are complementary and also 01 and 10 are complementary. By considering the complementary rule, we may use A, C, G and T to denote 00, 01, 10 and 11, respectively. This is an example of one possible representation and other representations are given in Table 1. An image pixel value (8-bits binary number) may be represented by four binary numbers each consisting of two bits. These numbers are mapped to their corresponding DNA sequence. Therefore, each pixel of the image is represented by four acid bases. The 8-bit pixel value can be encoded in 24 possible ways using 4 bases but Watson-Crick base pairing rules support only 8 representations

Table 1: DNA encoding and decoding rules.

Rule-1	Rule-2	Rule-3	Rule-4	Rule-5	Rule-6	Rule-7	Rule-8
00-A	00-A	00-T	00-T	00-C	00-C	00-G	00-G
01-C	01-G	01-C	01-G	01-A	01-T	01-A	01-T
10-G	10-C	10-G	10-C	10-T	10-A	10-T	10-A
11-T	11-T	11-A	11-A	11-G	11-G	11-C	11-C

as listed in Table 1. For example, if the decimal value of a pixel is 53, its 8-bit binary representation is 00110101, then it is encoded as a DNA sequence CGAA according to rule 5 in Table 1. Inversely, the pixel value can be decoded by DNA sequence, for example, DNA sequence TCTC is decoded as 10110111 by rule 7 and the corresponding decimal value is 187.

This paper has developed a dynamic DNA coding process in which the rules of Table 1 is selected according to a chaotic matrix, say A . The elements of A are integers in the range 1–8. The pre-processed image may be represented by another matrix, say B where the elements of B are integers in the range 1–255 for 8-bits binary representation of each pixel. The dimension of A and B are same (say 3×3 , for an example). Thus, the dimension of the DNA encoded matrix (say C) will be 3×12 , as illustrated below by taking arbitrary values of A and B .

$$A = \begin{pmatrix} 3 & 7 & 5 \\ 4 & 1 & 8 \\ 5 & 6 & 2 \end{pmatrix} \quad B = \begin{pmatrix} 78 & 157 & 246 \\ 124 & 42 & 207 \\ 19 & 127 & 213 \end{pmatrix}$$

$$C = \begin{pmatrix} C & T & A & G & T & A & C & A & G & G & T & C \\ G & A & A & T & A & G & G & G & C & G & C & C \\ C & A & C & G & T & G & G & G & T & G & G & G \end{pmatrix}$$

5. THE PROPOSED CRYPTOSYSTEM

A grey scale image $P(r \times c)$ is taken as input where r and c are the number of rows and columns, respectively. Conventionally, $P(i, j)$ represents the pixel value at position (i, j) . The proposed cryptosystem is divided into the following major steps: (i) generation of initial conditions

of Lorenz system from 32 character secret input key, (ii) Generation of chaotic key matrix R and reshaping the chaotic key matrix R to S (see step-2), (iii) DNA encoding and decoding. The stepwise image encryption procedure is described below:

Step-1: A user defined function takes 32 characters secret key and the dimension of the plain image as input and returns the chaotic key matrix and variables p and q . The substeps described below explain the working of the function. Initially, we set six variables as $m0 = 127, n0 = 127, p0 = 127, q0 = 127, p = 0$ and $q = 0$. the secret key is converted into a double precision matrix $k(1 \times 32)$ containing the ASCII value of each character. A loop is defined to run for eight times to generate the new values of the variables $m0, n0, p0$ and $q0$. Within the loop the following operations are done sequentially. A one-dimensional matrix t is constructed in which the element value ranges from 1 to 32.

Loop j

$Z = 9-j;$

$t = [(4 * j - 3)(4 * z)(4 * j - 2)(4 * z - 1)];$

$m = 10 + \text{mod}((k(t(1)) + k(t(2))), \text{abs}(k(t(3)) - k(t(4))));$

$n = 20 + \text{mod}((k(t(3)) + k(t(4))), \text{abs}(k(t(1)) - k(t(2))));$

$p = 10 + \text{mod}((m + n), \text{abs}(m - n));$

$q = 20 + \text{mod}((m + n), \text{abs}(n - m));$

$c = \text{mod}((m \oplus m0), 128);$

$d = \text{mod}((n \oplus n0), 128);$

$e = \text{mod}((p \oplus p0), 128);$

$f = \text{mod}(q \oplus q0, 128);$

$m0 = m0 + c;$

$n0 = n0 + d;$

$p0 = p0 + e;$

$q0 = q0 + f;$

End loop j.

Here, $t(x)$ denotes the element value of matrix t at position x , $\text{abs}(x)$ returns the absolute value of the number x , $k(t(x))$ denotes the element value of matrix k at position $t(x)$. The operator $\oplus$ indicates bitwise XOR operation. After the completion of the loop we redefine p and q as $p = m0 + p0$ and $q = n0 + q0$. We use the values of p and q to construct a one-dimensional matrix, $C(1 \times 6) = \text{linspace}(\frac{p}{1000}, \frac{q}{1000}, 6)$. Here, $\text{linspace}(a, b, n)$ returns a one-dimensional matrix of n linearly spaced elements ranging from a to b . Initially one dimensional matrix, $S0(1 \times 6) = [xyzuvw] = IC$, defines the six initial conditions for the Lorenz 6D system.

Step-2: Solve equation (1) by using the initial condition $S0$ and differential equation solver `ode45`. The iterative process has been executed $(n + 100)$ times and the results are stored in matrix $R((n + 100) \times 6)$; where $n = r \times c$. To obtain the steady state chaotic solution, we discard the sequence of first 100 iterations from the matrix R . The rest part of the R matrix is renamed as $S(n \times 6)$. To restrict the elements of S in the range 0–255 we have applied the following operation (2). The matrix $S(n \times 6)$ is stated below in equation (3).

$$S = \text{mod}((\text{round}(\text{abs}(S * 10^5))), 256). \quad (2)$$

$$S(n \times 6) = \begin{pmatrix} x_1 & y_1 & z_1 & u_1 & v_1 & w_1 \\ x_2 & y_2 & z_2 & u_2 & v_2 & w_2 \\ x_3 & y_3 & z_3 & u_3 & v_3 & w_3 \\ \dots & \dots & \dots & \dots & \dots & \dots \\ \dots & \dots & \dots & \dots & \dots & \dots \\ x_{n-2} & y_{n-2} & z_{n-2} & u_{n-2} & v_{n-2} & w_{n-2} \\ x_{n-1} & y_{n-1} & z_{n-1} & u_{n-1} & v_{n-1} & w_{n-1} \\ x_n & y_n & z_n & u_n & v_n & w_n \end{pmatrix} \quad (3)$$

The matrix S is used to produce six different chaotic key matrices K_i of dimension $r \times c$ by taking the consecutive elements of S rowwise. Here $i = 1, 2, 3, \dots, 6$.

Step-3: We pre-process the chaotic keys K_1 and K_2 according to the equation (4) to get the new random sequences K_{12} .

$$K_{12} = (K_1 \oplus K_2). \quad (4)$$

The next two chaotic sequences K_3 and K_4 are used to get four random sequences K_{r1}, K_{r2}, K_{r3} and K_{r4} of dimension $(r \times c)$ by the following Equations (5)–(8) and they are combined to obtain a matrix K_r of dimension $(r \times 4c)$

by putting them side by side according to equation (9).

$$K_{r1} = \text{mod}(K_3, 64) \quad (5)$$

$$K_{r2} = \text{mod}(K_4, 64) \quad (6)$$

$$K_{r3} = \text{mod}((K_3 + K_4), 64) \quad (7)$$

$$K_{r3} = \text{mod}(\text{abs}(K_3 - K_4), 64) \quad (8)$$

$$K_r = [K_{r1}K_{r2}K_{r3}K_{r4}]. \quad (9)$$

By using matrix K_r and equation (10) stated below, we get an binary array K_B of dimension $(r \times 4c)$.

$$K_B(i, j) = \begin{cases} 0, & \text{if } K_r(i, j) \geq 32 \\ 0, & \text{if } K_r(i, j) < 32 \end{cases} \quad (10)$$

Here, i and j vary from 1 to r and 1 to $4c$, respectively. The elements of the remaining two chaotic sequences K_5 and K_6 are converted into integers in the range 1–8 by the following Equations (11) and (12), respectively.

$$K_5 = \text{mod}(K_5, 8) + 1. \quad (11)$$

$$K_6 = \text{mod}(K_6, 8) + 1. \quad (12)$$

Step-4: The first two chaotic keys K_1 and K_2 are used for shuffling the pixel positions of the input image. The pixels of the plain image P are randomized by swapping the pixel positions between $P(K_1(i, j), K_2(i, j))$ and $P(K_2(i, j), K_1(i, j))$. Then the pixel shifted image matrix P and random sequences K_{12} are used to carry out the following operations (13)–(16) and get the matrix XP of dimension $r \times c$.

$$P_m = \frac{\sum_{i=2}^{r-1} \sum_{j=2}^{c-1} P(i, j)}{(r-2) \times (c-2)}, \quad (13)$$

$$XP(1, 1) = P(1, 1) \oplus K_{12}(1, 1) \oplus P_m, \quad (14)$$

$$XP(i, 1) = P(i, 1) \oplus K_{12}(i, 1) \oplus XP(i-1, c), \quad \text{for } i \neq 1 \quad (15)$$

$$XP(i, j) = P(i, j) \oplus K_{12}(i, j) \oplus XP(i, j-1), \quad \text{for } j \neq 1 \quad (16)$$

In Equations (15) and (16), i and j vary from 1 to r and 1 to c , respectively. By applying DNA encoding rule given in Table 1, encode matrix $XP(r, c)$ according to chaotic sequence K_5 to obtain DP of dimension $(r \times 4c)$ with the DNA elements A , C , G and T . Next the elements of DP are complemented according to the array K_B . If the

element of K_B is zero then the corresponding element of DP will remain unchanged but if the element of K_B is one then the corresponding element of DP will be replaced by its complement. Finally, the elements of DP are decoded according to chaotic sequence K_6 to get back the decimal values of the pixels of the encrypted image EP .

The decryption process is the reverse process of encryption. During decryption step-1 to step-3 of encryption process should be executed sequentially and step-4 should be executed in reverse order to get back the original plain image.

6. EXPERIMENTAL RESULTS AND SECURITY ANALYSIS

The 32 character secret key and different grey scale plain images of size $(r \times c)$ are taken as inputs and their encryption is done. For example, the plain images like, Lena, Barbara, and Cameraman along with their encrypted images, decrypted images with correct key $() + < > (7790n, [] \&^{\wedge} \% \$ \% \sim \text{tqpd}; 3b.l@kx$ and slightly different key (wrong key) $() + < > (7790n, [] \&^{\wedge} \% \$ \% \sim \text{tqpd}; 3b.l@ly$ are shown in Figure 2.

A cryptosystem is considered to be strong and secured enough if it is able to pass well-defined security tests. There exit a number of performance and security measurement techniques such as key space and sensitivity, histogram, correlation coefficient, information entropy, differential attack, noise attack, speed analysis etc. for benchmarking a cryptosystem. The various tests have been done and the results obtained are presented in this section with a comparison to the others algorithms reported in the literature recently.

6.1 Key space analysis

An inherent characteristic of a chaotic system is its high sensitivity to initial conditions. A slight change in initial condition may produce an entirely different sequence. In the present work, a novel use of 6D Lorenz system has been demonstrated. The 32 bit secret key goes through a series of modular arithmetic and bitwise-XOR operations. Only one bit change in the input key produces an entirely different chaotic sequence. The key space is 2^{256} which is sufficiently large to counter the brute force attack.

6.2 Key sensitivity analysis

To prove the validity of the proposed algorithm in terms of key sensitivity, we encrypt a plain image of Lena using two slightly different keys as stated below:

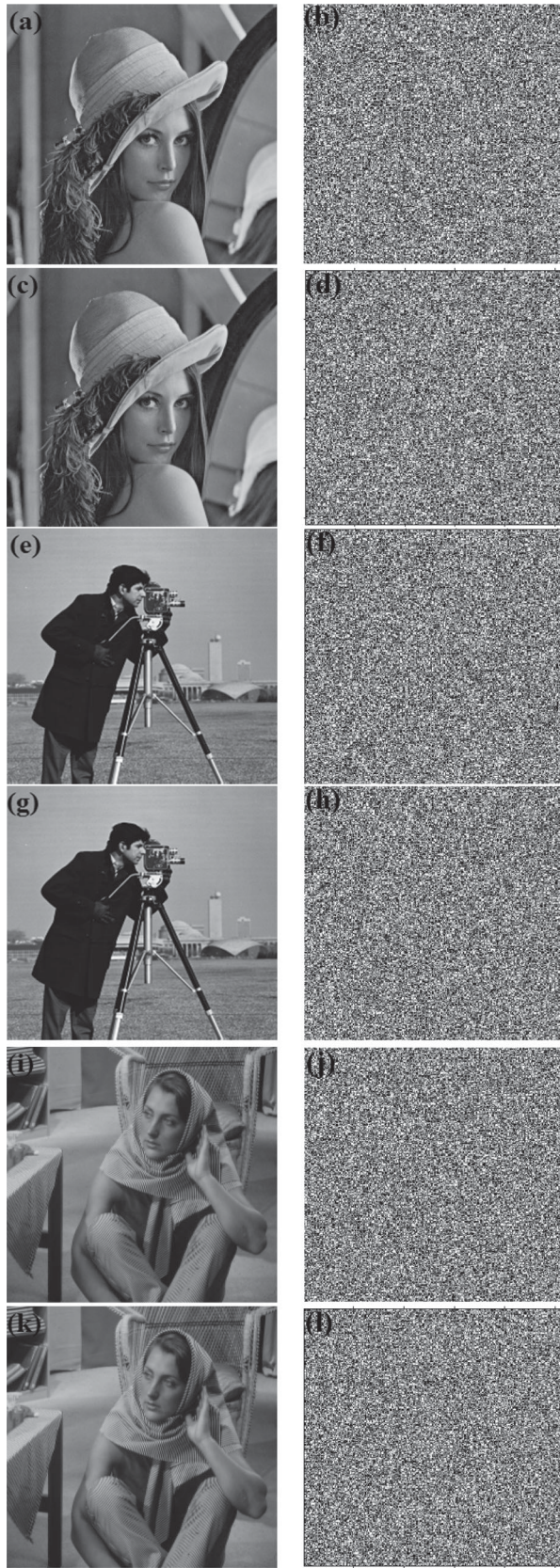


Figure 2: Plain images (Lena, Cameraman and Barbara) and their encryption and decryption results. Plain images (a), (e), (i); their corresponding encrypted images (b), (f), (j); decrypted images with correct key (c), (g), (k); decrypted images with wrong key (d), (h), (l).

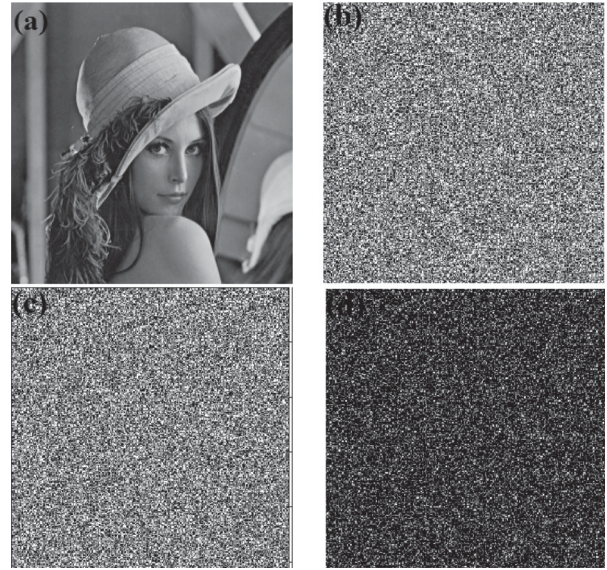


Figure 3: The results of key sensitivity test. (a) Plain image Lena, (b) Encrypted image with Key: $()+ < > (7790n,[]\&\^{\%}\$ \sim tqpd;3b.l@kx$, (c) Encrypted image with Key: $()+ < > (7790n,[]\&\^{\%}\$ \sim tqpd;3b.l@ky$, (d) Difference of (b) and (c).

Key-1: $()+ < > (7790n,[]\&\^{\%}\$ \sim tqpd;3b.l@kx$.

Key-2: $()+ < > (7790n,[]\&\^{\%}\$ \sim tqpd;3b.l@ly$.

Figure 3 describes the results of encrypting Lena image with slightly different keys (only single bit change between these two keys) and also the difference between two encrypted images. The encrypted images (Figure 3(b,c)) are remarkably different from one another because their difference is also chaotic in nature. This result proves that the encryption algorithm is highly sensitive to encryption key.

6.3 Histogram analysis

Histogram is an important tool for analysing statistical data. For an image, the histogram depicts how frequently different pixel values occur. The maxima and minima pixel value of an image falls in the range 0–255 for 8-bit representation. For a cryptographic algorithm to be secure, the encrypted image histogram should have a uniform distribution of pixel values so that no one can extract any information of the plain image from the histogram of the encrypted image. The histogram plot of the plain and the encrypted images are shown in Figure 4. The histogram of three different plain images (Figure 4(a,c,e)) are remarkably different whereas the histogram of their corresponding encrypted images (Figure 4(b,d,f)) are very much similar to one another.

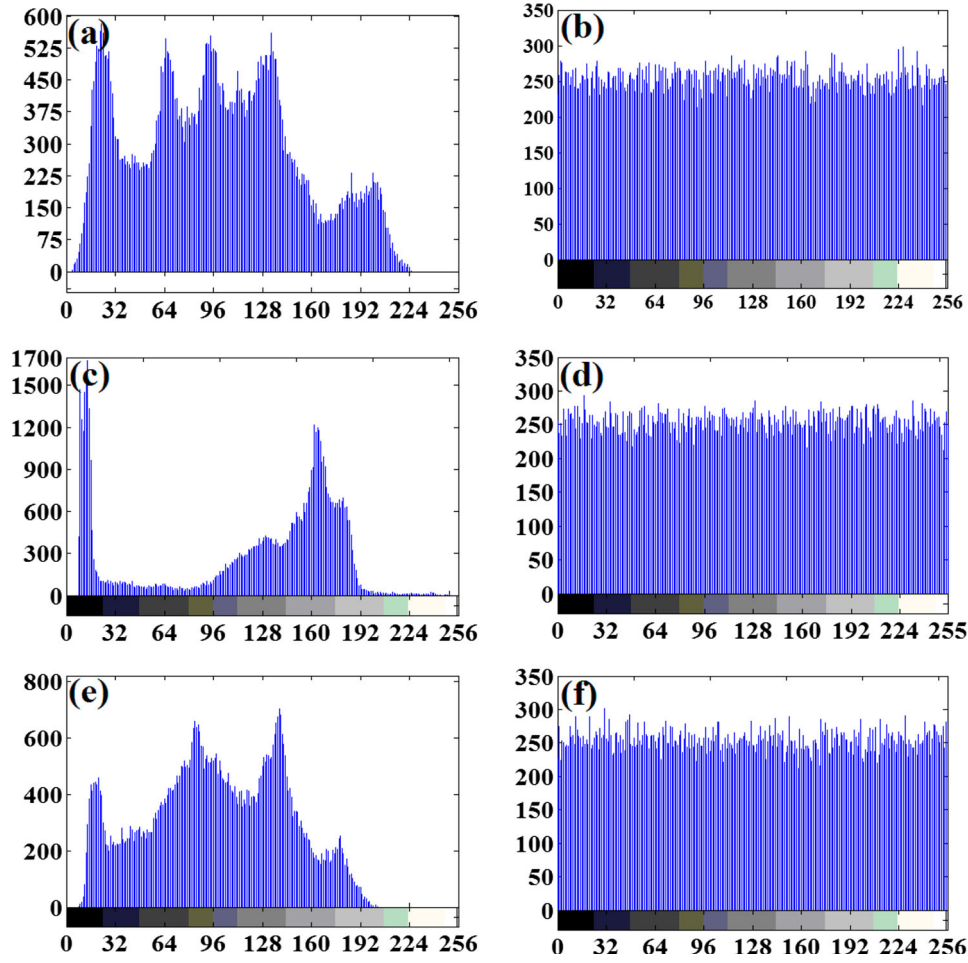


Figure 4: Histogram plot of various plain images and their encrypted version. (a) Plain Lena, (b) Encrypted Lena, (c) Plain Cameraman, (d) Encrypted Cameraman, (e) Plain Barbara, (f) Encrypted Barbara.

These results confirm that the encryption algorithm is highly secure against histogram matching and analysis based attack.

6.4 Correlation analysis

Another tool for measuring the security performance of a cryptosystem is correlation analysis. It is obvious that the pixel correlation of plain image is high and this value decreases sharply for encrypted images. To find out pixel correlation, we take two adjacent pixels (in horizontal, vertical and diagonal directions) from different places of the plain and encrypted images randomly. Then the pixel correlations are measured in these three directions. In this experiment, we have taken 3500 pairs of adjacent pixels randomly in horizontal, vertical and diagonal directions from different positions of both the plain and encrypted images. The computed values of the correlation coefficient applying the formulas stated below are given in Table 2. The corresponding scattered plots are shown in Figure 5. Table 3 presents a comparative

Table 2: Correlation coefficients of various test images and their encrypted counterparts.

Images	Horizontal	Vertical	Diagonal
Plain Lena	0.9400	0.9693	0.9179
Encrypted Lena	0.0023	0.0013	-0.0067
Plain Cameraman	0.9335	0.9592	0.9087
Encrypted Cameraman	0.0069	0.0003	0.0052
Plain Barbara	0.9029	0.9400	0.8848
Encrypted Barbara	0.0031	0.0014	-0.0016

analysis of correlation coefficients obtained from our proposed method with other methods reported in the recent literature.

$$E_x = \frac{1}{N} \sum_{i=1}^N x_i, \quad (17)$$

$$D_x = \frac{1}{N} \sum_{i=1}^N (x_i - E(x))^2, \quad (18)$$

$$cov(x, y) = \frac{1}{N} \sum_{i=1}^N (x_i - E(x))(y_i - E(y)). \quad (19)$$

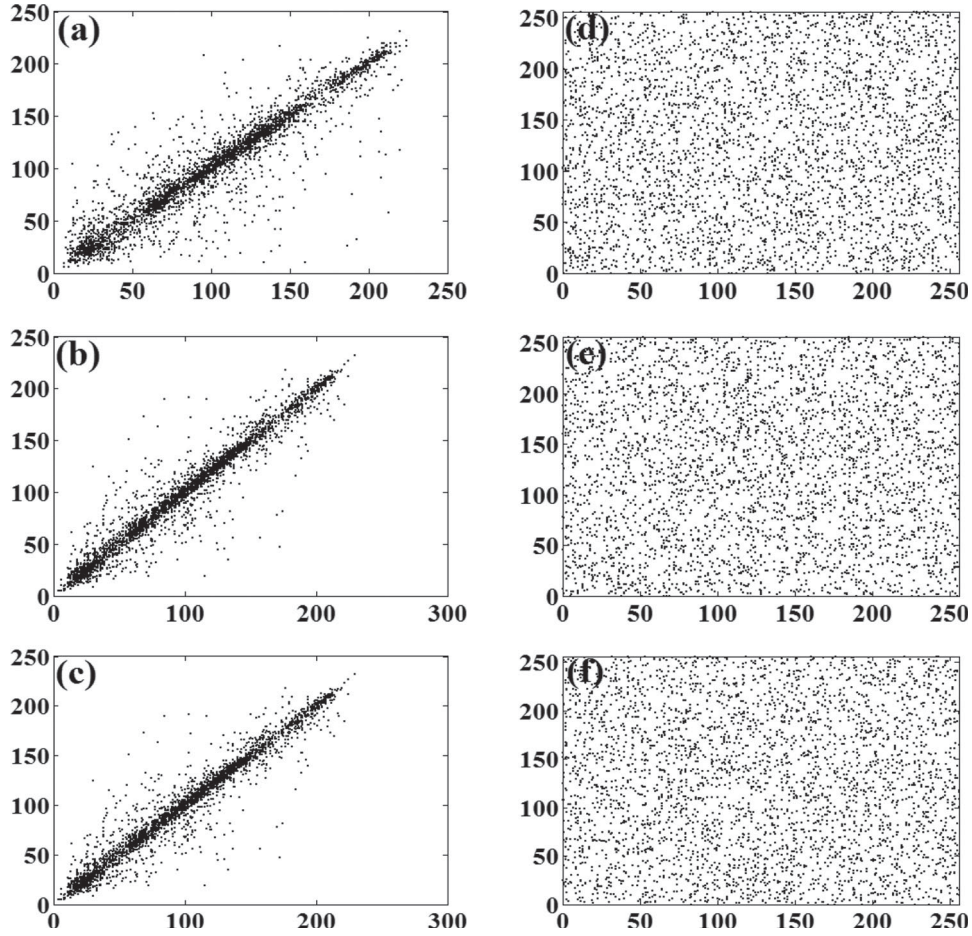


Figure 5: Correlation among two adjacent pixels. (a) Plain Lena in horizontal direction, (b) Plain Lena in vertical direction, (c) Plain Lena in diagonal direction, (d) Encrypted Lena in horizontal direction, (e) Encrypted Lena in vertical direction, (f) Encrypted Lena in diagonal direction.

Table 3: Comparative studies of correlation coefficients of encrypted Lena.

Encryption algorithm	Horizontal	Vertical	Diagonal
Ref. [3]	-0.043970	-0.003190	-0.008500
Ref. [4]	-0.0013	-0.0274	-0.0199
Ref. [8]	0.0021	0.0046	0.0033
Ref. [10]	0.0214	0.0465	-0.0090
Ref. [12]	0.0011	0.0040	0.0080
Ref. [13]	0.0796	0.0166	0.0032
Ref. [15]	-0.0029	0.0015	-0.0076
Ref. [18]	-0.0230	0.0019	-0.0034
Proposed algorithm	0.0023	0.0013	-0.0067

$$r_{xy} = \text{cov}(x, y) / (\sqrt{D_x}) (\sqrt{D_y}), \quad (20)$$

Here x, y represent the adjacent grey scale pixel values of the test image. N denotes the total number of duplets (x, y) obtained from the test image. For a good cryptosystem, the correlation coefficients for the encrypted image should be very small, *i.e.* close to zero.

6.5 Information entropy analysis

Information entropy is a method of measuring the degree of randomness of an information source. The pixel value of an image falls in the range 0–255. Hence it has 256 states and the maximum entropy would ideally be 8 which is possible if all the pixels have different grey levels. The information entropy, m is measured by using the formula given below:

$$E(m) = \sum_{i=0}^{2^N-1} P(m_i) \log_2 \frac{1}{P(m_i)} \quad (21)$$

where $P(m_i)$ is the probability of symbol m_i and each symbol is represented by N bits. Table 4 gives the results of entropy analysis of grey scale test images. It is observed that the entropy reaches towards the ideal value 8. Table 5 compares the information entropy of our proposed method with the other recently proposed method.

Table 4: Information entropy of plain and encrypted images.

Images	Plain image	Encrypted image
Lena	7.5690	7.9975
Cameraman	7.0097	7.9974
Barbara	7.3975	7.9972

Table 5: Comparative studies of information entropy.

Encryption algorithm	Encrypted Lena	Encrypted Cameraman
Ref. [3]	7.989020	–
Ref. [9]	7.9970	7.9967
Ref. [12]	7.9970	7.9972
Ref. [14]	7.9895	–
Ref. [20]	7.9974	–
Ref. [21]	7.9826	7.9637
Proposed algorithm	7.9975	7.9974

6.6 Differential attack analysis

The differential attack is a technique of finding the relation between two encrypted images making a small variation in the plain image. For a good cryptosystem, a small variation in input image should generate a totally different encrypted image. Two metrics: Number of Pixels Change Rate (NPCR) and Unified Average Changing Intensity (UACI) are used to analyse the strength of an encryption algorithm against differential attack. NPCR is defined as the percentage of the number of different pixels at the same positions between two encrypted images whose plane images are of only one pixel difference. UACI indicates the average intensity of differences between two encrypted images with one pixel change in their plane images. Let us consider two encrypted images E_1 and E_2 and their corresponding plain images differ only by one pixel. Let $E_1(i, j)$ and $E_2(i, j)$ are the pixel values at location (i, j) of E_1 and E_2 , respectively. Now, we construct a bipolar array D with elements 0 or 1 using equation (22). The dimension of D is same as the dimensions of E_1 and E_2 .

$$D(i, j) = \begin{cases} 1, & \text{if } E_1(i, j) \neq E_2(i, j) \\ 0, & \text{Otherwise} \end{cases} \quad (22)$$

The NPCR and UACI are computed by applying the following formulas (23)–(24).

$$\text{NPCR} = \frac{\sum_{i,j} D(i, j)}{r \times c} \times 100\%, \quad (23)$$

$$\text{UACI} = \frac{1}{r \times c} \left[\frac{\sum_{i,j} |E_1(i, j) - E_2(i, j)|}{2^N - 1} \right] \times 100\%. \quad (24)$$

NPCR and UACI metrics are computed on various plain images and the results are tabulated in Table 6. A comparative study of different recent encryption systems has been displayed in Table 7 based on the values of NPCR and UACI. The theoretically expected values of NPCR

Table 6: NPCR and UACI values for one pixel difference in a plain image.

Images	Pixel positions	Changed value of pixel	NPCR	UACI
Lena	(1,1)	137 → 138	99.63	33.42
	(17, 243)	21 → 22	99.62	33.52
	(100,200)	25 → 26	99.63	33.56
	(130,11)	56 → 55	99.60	33.46
	(200,210)	73 → 74	99.63	33.39
	(256,256)	72 → 73	99.66	33.24
	Mean value		99.63	33.43
Cameraman	(1,1)	156 → 157	99.60	33.34
	(17, 243)	155 → 156	99.60	33.50
	(100,200)	162 → 163	99.60	33.61
	(130,11)	180 → 181	99.60	33.52
	(200,210)	133 → 134	99.64	33.46
	(256,256)	113 → 114	99.56	33.47
	Mean value		99.60	33.48
Barbara	(1,1)	85 → 86	99.60	33.58
	(17, 243)	10 → 11	99.60	33.41
	(100,200)	120 → 121	99.60	33.38
	(130,11)	184 → 185	99.61	33.50
	(200,210)	27 → 28	99.57	33.45
	(256,256)	89 → 88	99.62	33.48
	Mean value		99.60	33.47

Table 7: Comparison of NPCR and UACI for encrypted Lena.

Encryption algorithm	NPCR	UACI
Ref. [4]	99.61	33.46
Ref. [5]	99.62	27.38
Ref. [8]	99.63	33.71
Ref. [11]	99.60	33.44
Ref. [14]	99.65	33.34
Ref. [15]	99.62	33.06
Ref. [16]	99.62	33.51
Ref. [17]	99.61	33.50
Ref. [20]	99.62	33.51
Ref. [21]	99.61	33.33
Proposed method	99.64	33.56

and UACI [22,23] for two true random images may be expressed by the Equations of (25) and (26). For $N = 8$, $\text{NPCR}_{\text{expd}} = 99.609\%$ and $\text{UACI}_{\text{expd}} = 33.464\%$.

$$\text{NPCR}_{\text{expd}} = \left(1 - \frac{1}{2^N} \right) \times 100\% \quad (25)$$

$$\text{UACI}_{\text{expd}} = \frac{1}{2^{2N}} \frac{\sum_{i=1}^{2^N-1} i(i+1)}{2^N - 1} \times 100\%. \quad (26)$$

6.7 Randomness test

The verification of randomness of a set of digital numbers is essential for demanding it as a chaotic sequence. The National Institute of Standards and Technology (NIST) Commission has released the random and pseudo-random sequence test standard SP 800–22 on April 2010 [24]. This test standard consists of 16 different tests and each test will produce a p -value as the result. If the p -value $\geq \alpha$ (In SP 800–22 test $\alpha = 0.01$) then we can say that the digital sequence passed the randomness test. A

Table 8: SP 800–22 random and pseudo-random sequence test.

Test name		Chaotic key		Cipher image	
		<i>p</i> -Value	Remarks	<i>p</i> -Value	Remarks
Frequency		0.3154	Success	0.3845	Success
Block frequency (<i>M</i> = 128)		0.5113	Success	0.4615	Success
Runs		0.3523	Success	0.4529	Success
Longest runs of one (<i>M</i> = 128, <i>N</i> = 49, <i>K</i> = 5)		0.7632	Success	0.2615	Success
Rank		0.4224	Success	0.3745	Success
Spectral DFT		0.4253	Success	0.7381	Success
Non overlapping templates (<i>m</i> = 9, <i>B</i> = 101000111)		0.5527	Success	0.6642	Success
Overlapping templates (<i>M</i> = 1032, <i>N</i> = 508)		0.1637	Success	0.3427	Success
Universal (<i>L</i> = 6, <i>Q</i> = 640, <i>K</i> = 86741)		0.8932	Success	0.9115	Success
Linear complexity (<i>M</i> = 500)		0.7328	Success	0.6917	Success
Serial (<i>m</i> = 10)	<i>p</i> -Value 1	0.8632	Success	0.8825	Success
	<i>p</i> -Value 2	0.8315	Success	0.8937	Success
Approximate entropy (<i>m</i> = 10)		0.6652	Success	0.7268	Success
Cumulative sums	Forward	0.3526	Success	0.3647	Success
	Reverse	0.3862	Success	0.2973	Success
Random excursions (<i>x</i> = 1)		0.9581	Success	0.8385	Success
Random excursions variant (<i>x</i> = 1)		0.7029	Success	0.6572	Success

chaotic sequence has good randomness if it passes all the tests. Table 8 shows the results of SP 800–22 test for the chaotic key and the encrypted image. The total number of bits of this system are 524288 by considering a 256×256 image in grey scale representation.

6.8 Speed analysis

The encryption time is also considered as a performance parameter of a cryptosystem. The cryptosystem should be as quick as possible in encrypting and decrypting images. Our cryptosystem basically consists of XOR operations, modular operations and arithmetic operations like multiplication, addition, subtraction to reduce the execution time as far as possible. The algorithm has been tested and designed in MATLAB R2009a running on an Intel(R) Core(TM) i3–5010U CPU @ 2.10GHz, 4GB memory with Windows 10 Home Single Language 64-bit operating system. Table 9 lists the encryption times taken by our cryptosystem tested on various plain images. The encryption algorithm is run 10 times for each image and the mean execution time is calculated and given in Table 9.

Table 9: Encryption time of various test images.

Test image	Dimension	Encryption time
Lena	256×256	0.1641 s
Cameraman	256×256	0.1745 s
Barbara	256×256	0.1631 s

7. CONCLUSION

A novel image encryption algorithm using 6D hyperchaotic Lorenz system and DNA encoding/decoding technique is presented in this paper. The proposed encryption system is fast and highly secured. The security against different attacks is analysed. It is seen that the ciphered image is purely dependent on the plain image pixel, previous ciphered pixel, chaotic key and DNA encoding/decoding rule. A single bit change in the key or the plain image produces an entirely different encrypted image. The time taken for encryption is less than one second which is good enough. The results obtained from different security tests are also very good and detailed comparison with other algorithms reported in the literature is also done. The proposed cryptosystem can be implemented for practical encryption applications.

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